CaffeinatedData

Two Sample T-Test Project

Research Question:
Do customers who churn have a higher monthly charge compared to customers who stay? This can be answered with a two-sample t-test.

If churned customers are paying higher monthly charges on average, it could indicate that competitors have better prices and the company's pricing is the driver of churn. This investigation would help determine if it is worth giving discounts or bundled services to entice customers to be loyal. It is more difficult to gain customers than to retain loyal customers.

"*Given that it costs 10 times more to acquire a new customer than to retain an existing one, customer retention has now become even more important than customer acquisition*."

The variables that are relevant to the question would be:

Independent Variable: Churn (Categorical variable, "Yes" and "No" values)

Dependent Variable: MonthlyCharges (Continuous variable, Monthly dollar amount billed to each customer)

Business Problem:

The company wants to determine whether customers who churn pay higher monthly charges than retained customers. If pricing is associated with churn, pricing strategies or bundled services may improve customer retention.

Data Analysis Process:

I used a two-sample T-test in order to conduct this hypothesis testing because I am comparing a continuous variable (MonthlyCharge) and a categorical variable (Churn Yes/No) for the difference of means.
A Chi square would not be applicable since Chi square is for only categorical variables and Anova is for 3 groups. A significance level of 0.05 will be chosen.

MonthlyCharge is the dependent variable and Churn is the independent variable since the hypothesis is do churners pay more than non-churners?

The null hypothesis is that churners do not pay more than non-churners.

The alternative hypothesis is that churners do pay more than non-churners.

The test is right-tailed.

Code:

```
#We need the mean, standard deviation, and number of observations for each group in order to fill in the formula for the Two sample T test

xbar = df.groupby('Churn')['MonthlyCharge'].mean()

print("means :", xbar)

s= df.groupby('Churn')['MonthlyCharge'].std()

print('standard deviation : ', s)

n = df.groupby('Churn')['MonthlyCharge'].count()

print('count of values : ', n)

varxy =df.groupby('Churn')['MonthlyCharge'].var()

print('Variance :', varxy)

print('Variance ratio' ,1546/1703 , 1703/1546)


#Assigning values to the variables in order to plug into the formula

#means of the group
```

xYes = 199.295175

xNo = 163.008973

#Standard deviation

sYes = 41.268191

sNo = 39.322148

#number of variables

nYes = 2650

nNo = 7350

T test calculation

numerator = xYes - xNo

denominator = np.sqrt(sYes**2/nYes + sNo **2/ nNo)

t_stat = numerator / denominator

print(t_stat)

#observed difference between churners and non-churners in Monthlycharge as a T statistic is **$39.28**

#observed difference between churners and non-churners in MonthlyCharge as group **means** is **$36.29**

#Calculating degrees of freedom

dofd = nYes + nNo - 2

print(dofd)

pvalue = 1- t.cdf(t_stat, df = dofd)

print(pvalue)

Means of both groups:

Churn = “No”: 163.008973

Churn = “Yes”: 199.295175

On average, customers who churned paid about $36 dollars more per month than those who did not churn .

Mean difference: $36.29

Standard Deviation of both groups:

Churn = “No”: 39.322148

Churn = “Yes”: 41.268191

The spread between the two groups is similar which suggests variability is not drastically different.

Count of values:

Churn = “No”: 7350

Churn = “Yes”: 2650

Both groups have large sample sizes which makes the T-Test more reliable.

Variance of both groups:

Churn = “No”: 1546.231296

Churn = “Yes”: 1703.063567

Variance Ratio:

1546/1703= 0.907809747504404

Since the variance ratio is close to 1, the two groups are strongly homogeneous.

T Statistic:

39.28778554725533

The large t-value shows that the difference between churners and non-churners is significant.

Degrees of freedom:

9998

P Value:

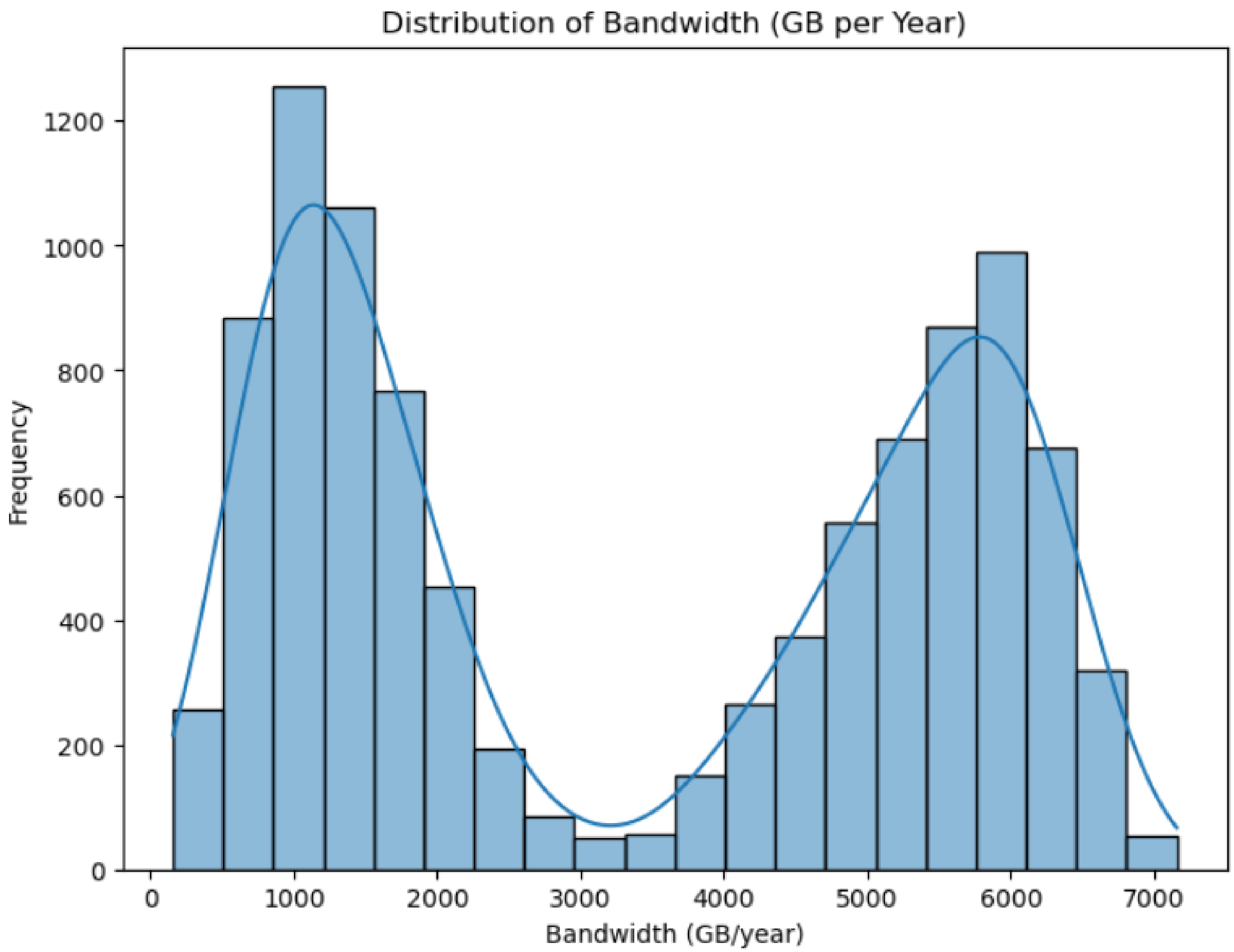
0.0

The P value is extremely small.

The two sample t-test was the best choice because:

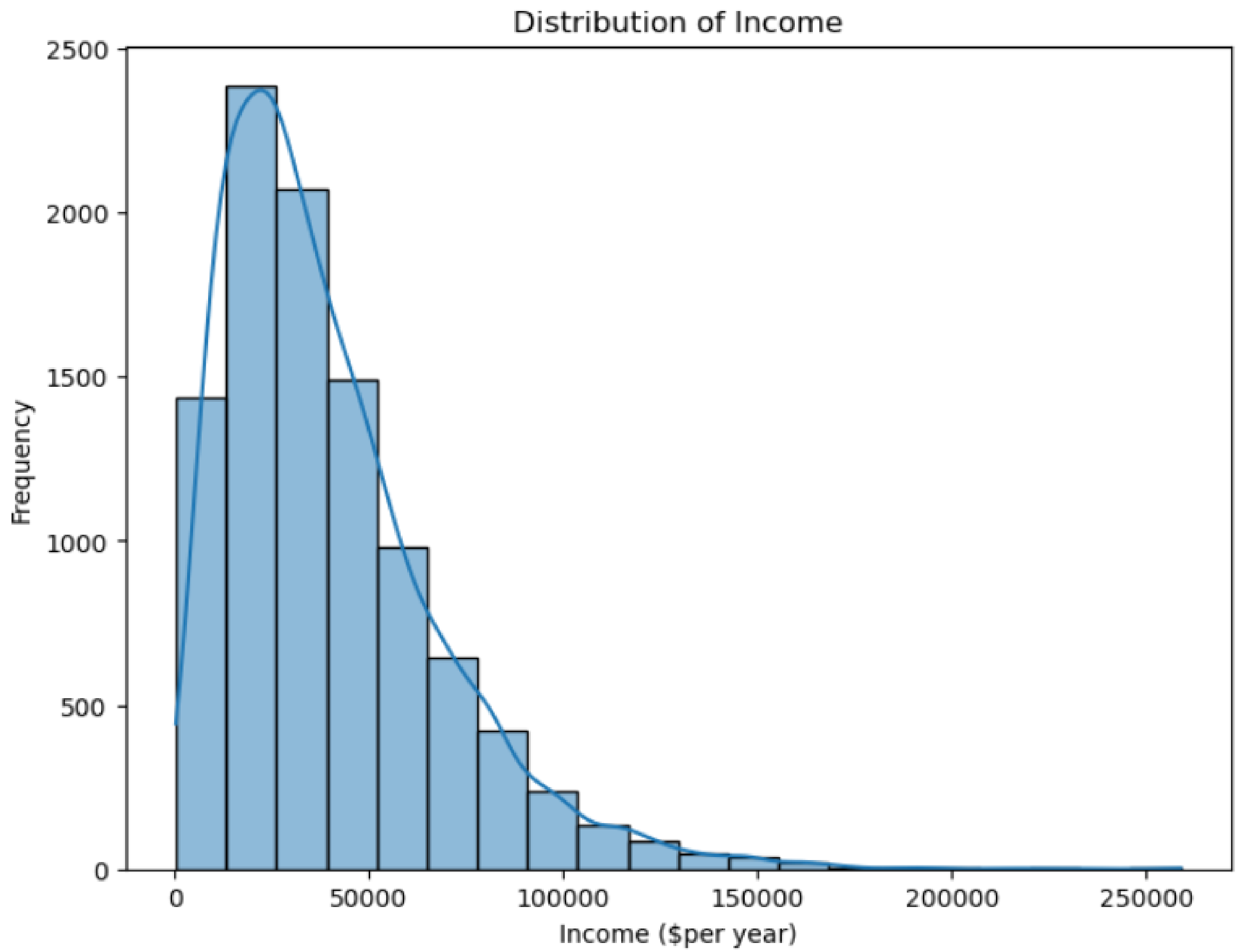
1. The research question compares the mean of a continuous outcome (**MonthlyCharge**) between two independent groups (Churn: “Yes” vs. “No) which matches a two-sample t-test
2. A chi-square test is inappropriate because it only analyzes categorical vs. categorical, not differences in continuous means.
3. An ANOVA test is not necessary because it is designed for three or more groups. This study only involves two groups.
4. The variance of the two groups are similar (the ratio is close to 1, **variance ratio = 0.91**),
5. With large sample sizes (7350 for “No” and 2650 for “Yes”), the Central Limit Theorem supports normality of the sampling distributions of the mean difference.

Continuous Variable "Bandwidth_gb_year"



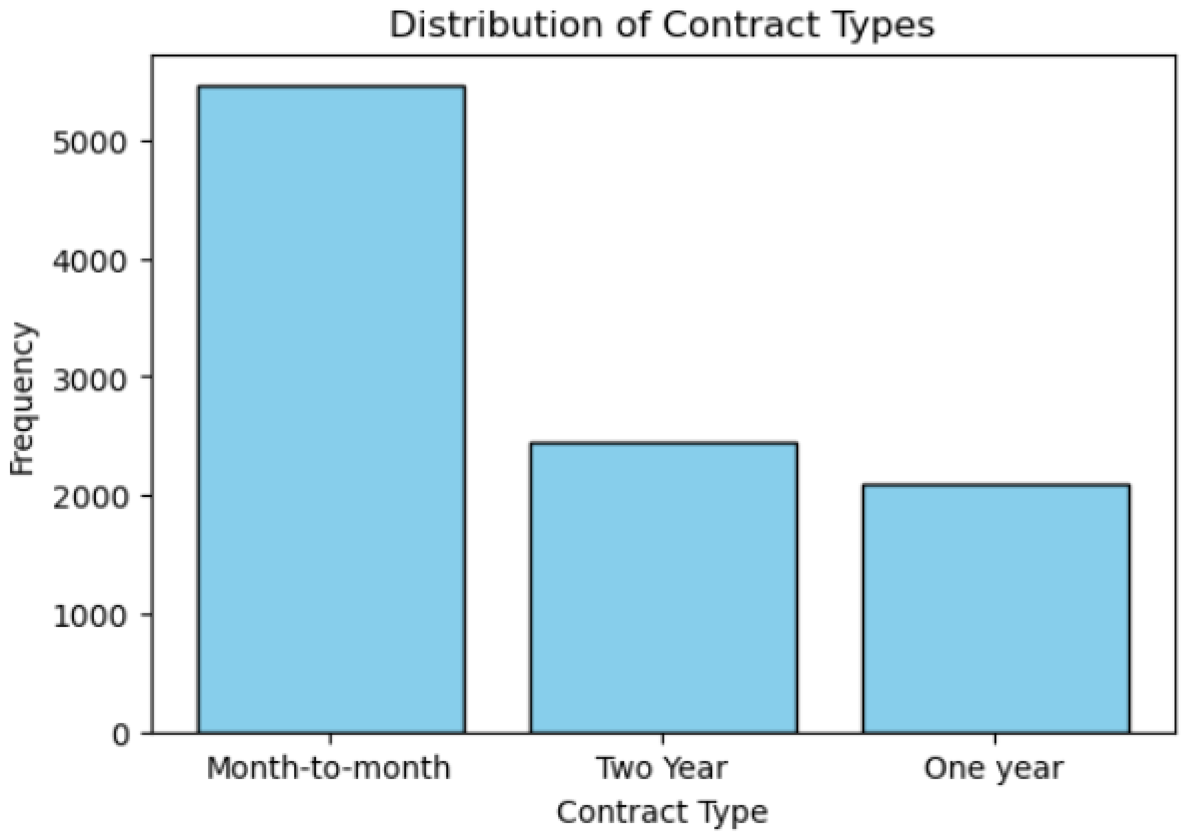
Bandwidth_GB_Year	Value
Count	10000
Mean	3392.34
Median	3279.53
Min	155.5
Q1 (25%)	1236.47
Q3 (75%)	5586.14
Max	7158.98
Range	7003.47
STD.Deviation	2185.29

Continuous Variable: "Income"



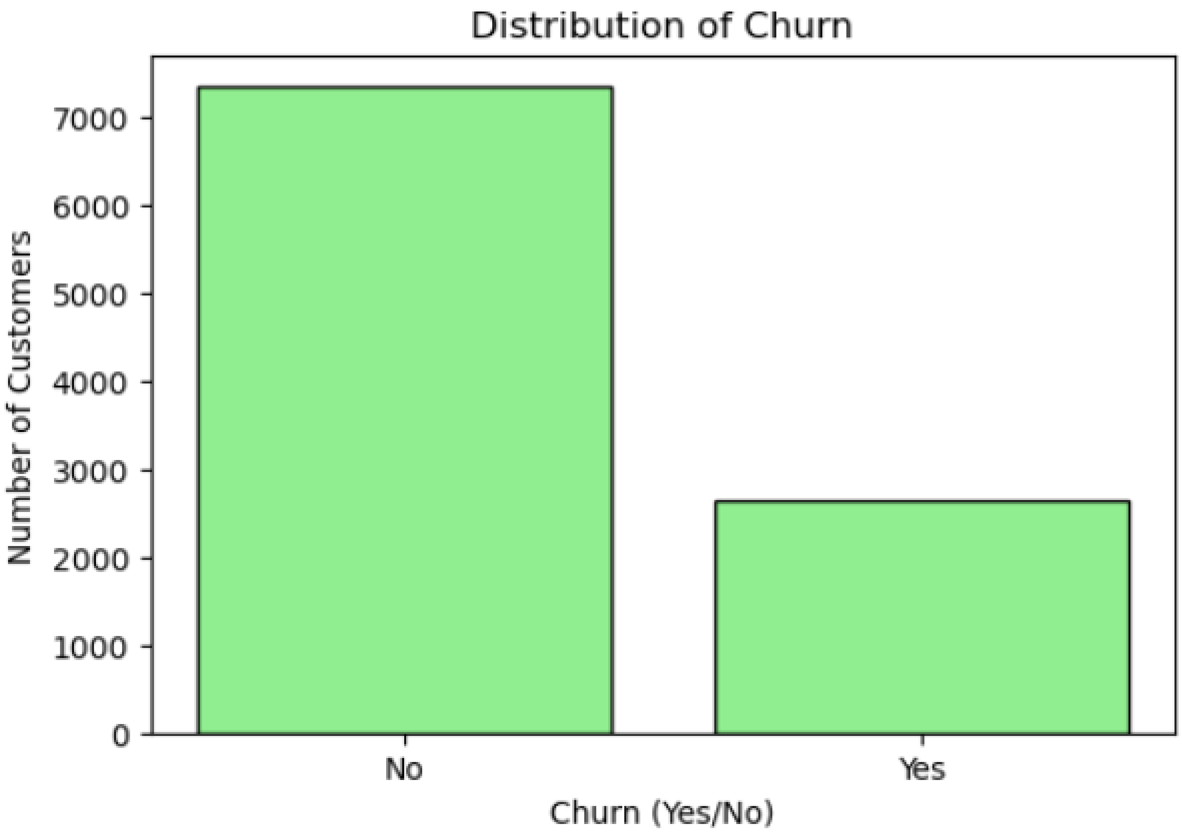
Income	Value
Count	10000
Mean	39806.92
Median	33170.6
Min	348.67
Q1 (25%)	19224.71
Q3 (75%)	53246.17
Max	258900.7
Range	258552.03
STD.Deviation	28199.91

Categorical Variable: “Contract”



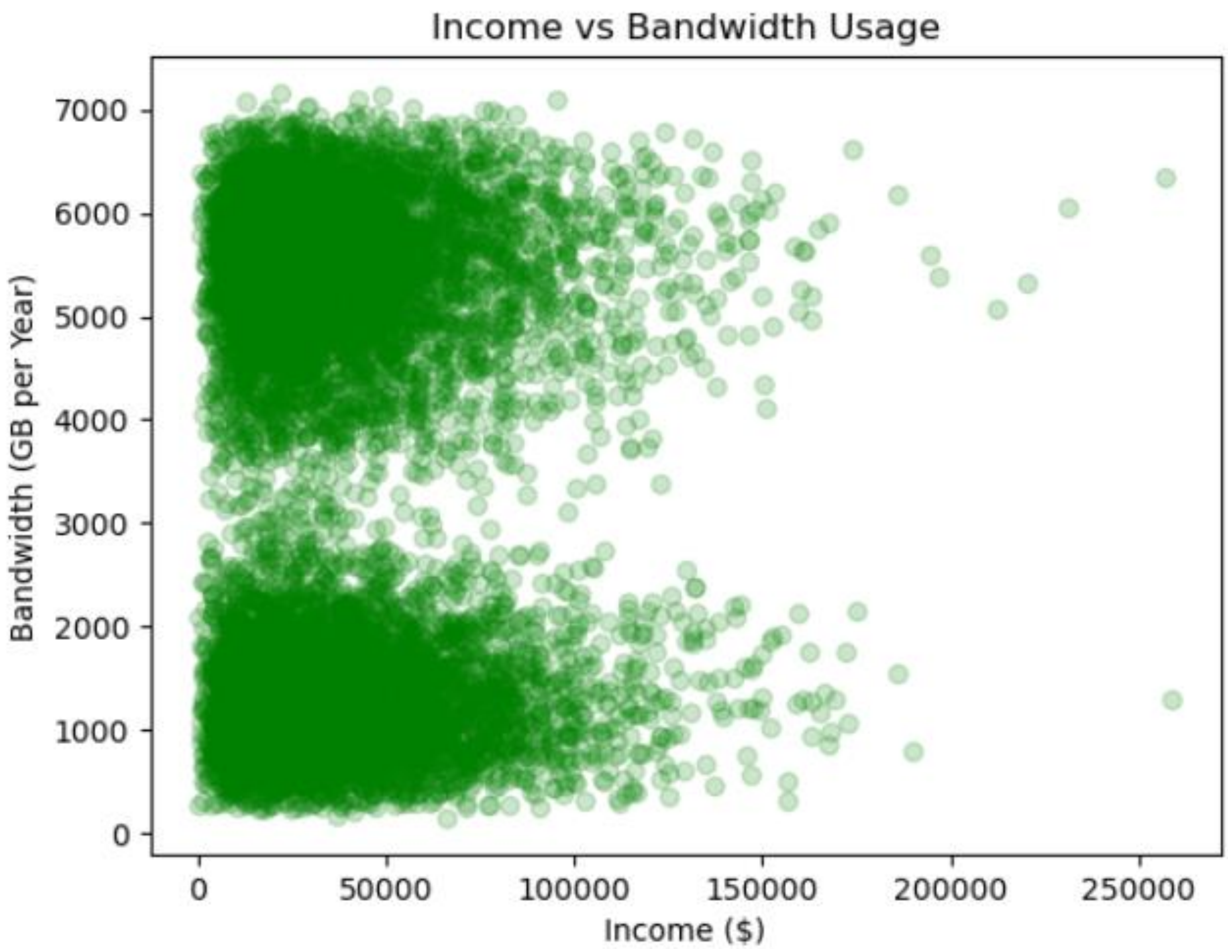
Contract (categorical)	Values
count	10000
"Month to Month"	54.55%
"One Year"	24.42%
"Two Year"	21.02%

Categorical Variable: “Churn”



Churn (categorical)	Values
count	10000
"No"	73.50%
"Yes"	26.50%

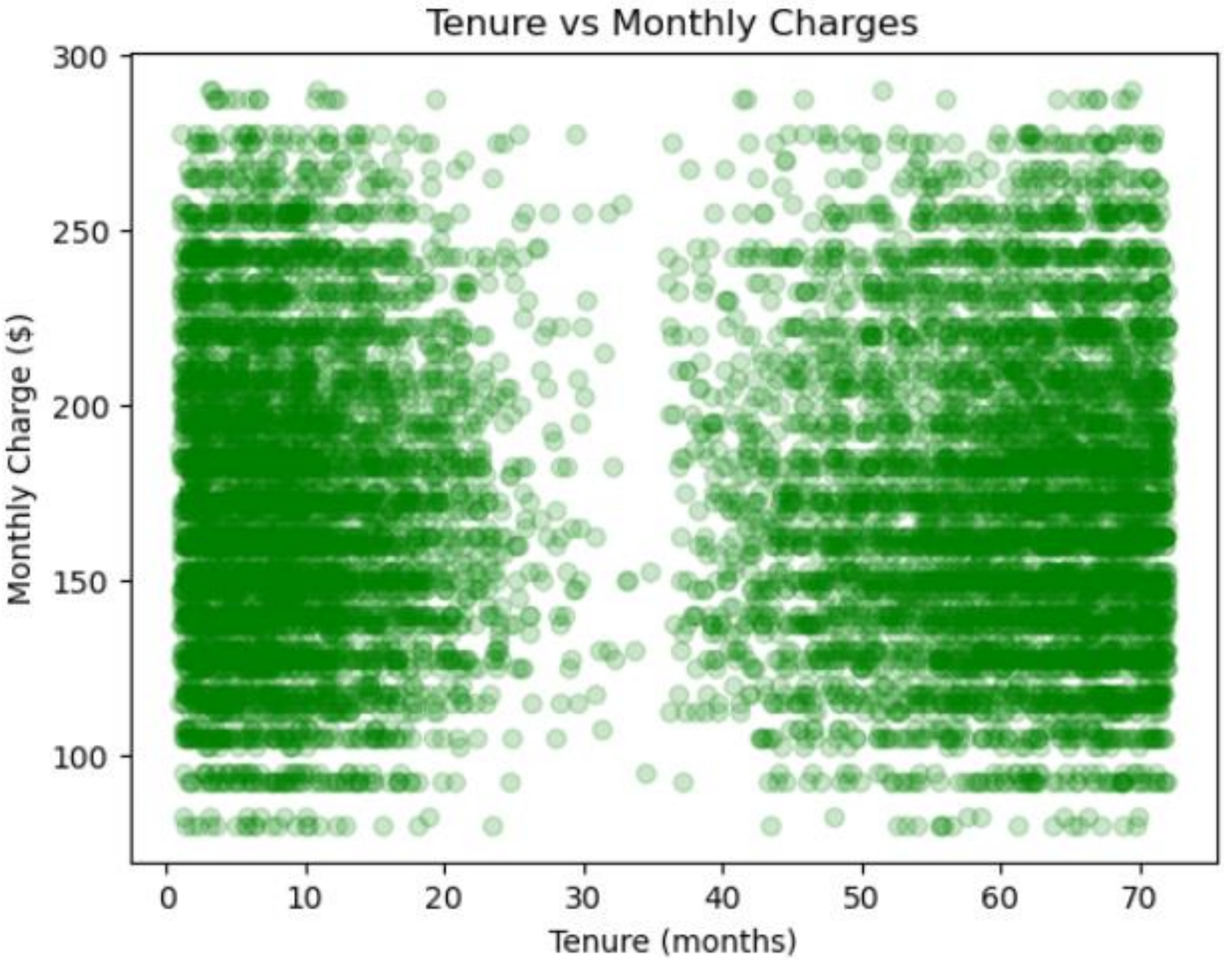
Continuous variable bivariate Scatterplot with “Income” and “Bandwidth_GB_Year”



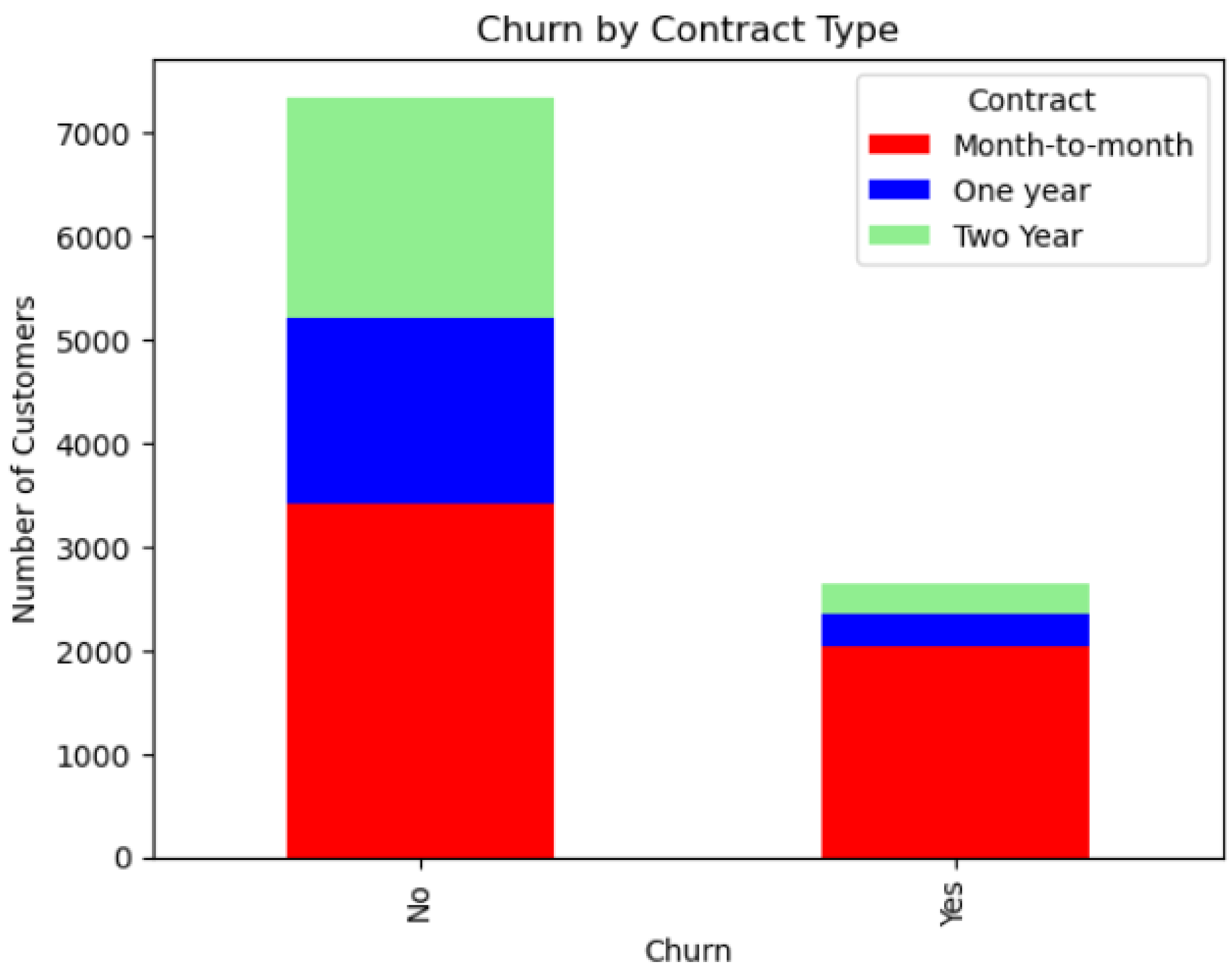

```
[453]: #Correlation between the two continous variables "Income" and "Bandwidth_GB_Year"
       r = df['Income'].corr(df['Bandwidth_GB_Year'], method='pearson')
       print(r)

       0.003673549628484681
```

The Pearson’s correlation coefficient is **0.00367.** The scale range is from -1 to 1 so therefore this value indicates that the two variables “Income” and “Bandwidth_Usage” have almost no correlation.

Continuous variable bivariate Scatterplot with “Tenure” and “MonthlyCharge”



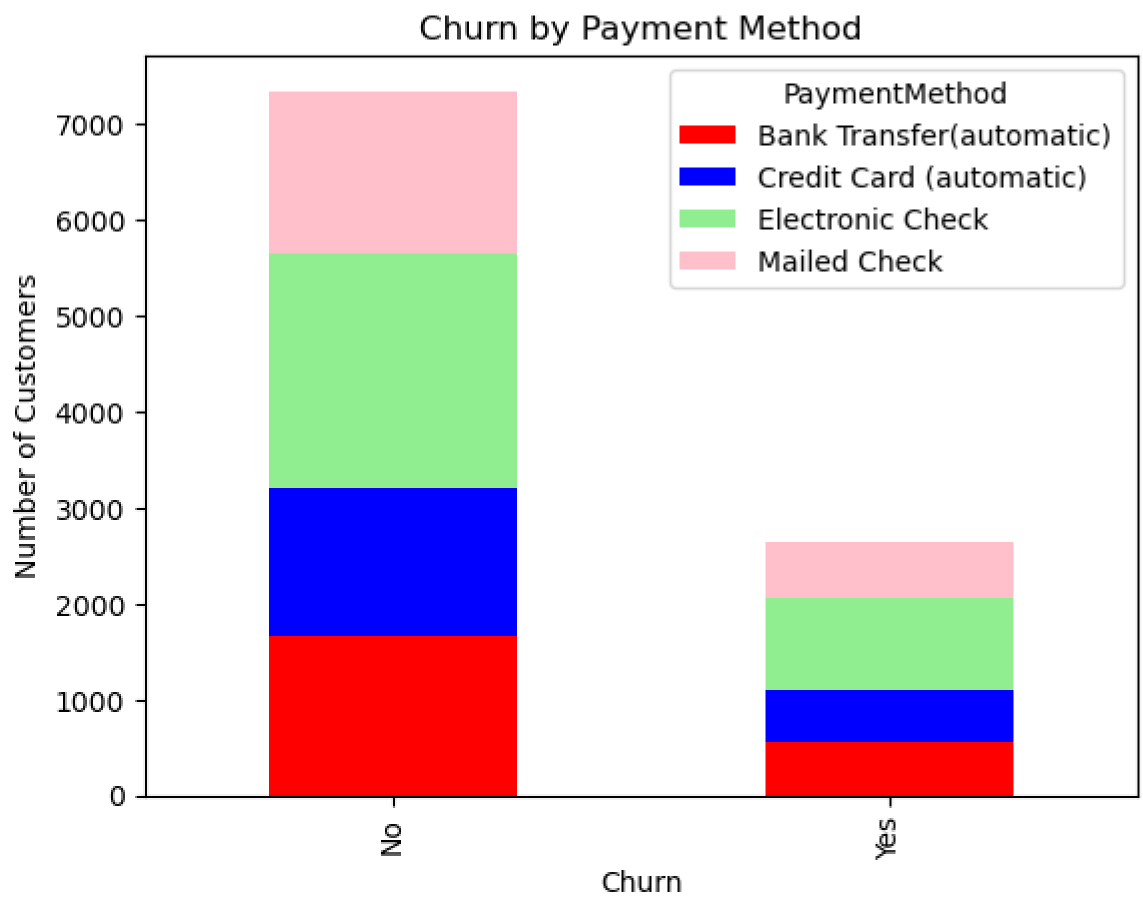
Categorical Variable Bivariate crosstab graph “Churn” and “Contract”:




```
[492]: #Cramers V association test for two categoricals

       table = pd.crosstab(df['Churn'], df['Contract'])
       cramers_v = association(table, method="cramer")
       print("Cramér's V:", cramers_v)

       Cramér's V: 0.26806558536950914
```

There is a moderate association between the two variables “Churn” and “Contract” with the value **0.268**. The scale is from 0 to 1 so we can treat it as 26**%**/27% association. As shown in the stacked bar chart. Customers in a **Two-year** contract have the least churn by percentage while customers in a **month to month** contract have the most churn.

Categorical Variable Bivariate crosstab graph “Churn” and “PaymentMethod”:



Conclusion:

I tested where customers who churn have higher monthly charges than those who stay using a right-tailed two sample t-test (similar variance, significance level = 0.05)

The null hypothesis is that churners do not pay more than non-churners

The alt hypothesis is that churners do pay more than non-churners.

Mean difference between the two groups: $36.29

The test produced a t-statistic of approximately **39.28**, and a p-value close to **0.000**, well below the significance level threshold of **0.05**. The results provide strong evidence to reject the null hypothesis. Therefore, churned customers **do pay significantly more on average** than non-churned customers. This suggests that higher monthly charges are strongly associated with customer churn

White the dataset for “MonthlyCharge” is large and satisfies Central Limit Theorem, it is not truly a random sample. This may limit how the results can be generalized. Additionally, the t-test assumes homogeneity of variance, which can only be approximated. Though calculating the variance it comes close to 1 to 1. The variance ratio was 0.9.

T-test also only shows association and not truly the real cause of churn, it shows just one aspect of the concern. Other factors such as service issues like equipment failure and outages, location, contract type, better service from other competitors could be a factor of churn.

Recommendations:

Based on the findings, the company should consider strategies to reduce churn among high-paying customers. These may include offering discounts, promotions, or loyalty programs targeted at customers with higher monthly charges. Additionally, encouraging customers to adopt longer term contracts could reduce churn rates, as churn was far lower for two year contract customers as visualized in one of the categorical bivariate graphs.